פרויקט מס׳

BS-CS-26-197

Entity-Centric Analysis of Political Polarization:

Tracking the Israeli Judicial Reform Discourse on X

פרויקט מחקרי

דוח מסכם

הוכן לשם השלמת הדרישות לקבלת

תואר ראשון במדעי המחשב (B.Sc.)

מאת

אדגר אברמוב ושלי גולדמן

בהנחיית דר. אביעד אלישר

הוגש למחלקה למדעי המחשב

המכללה האקדמית להנדסה סמי שמעון

באר-שבע

תאריך עברי	תאריך לועזי
י״ב באלול התשפ״ו	25/08/2026

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Abstract:

This project investigates the digital polarization surrounding the Israeli judicial reform through a novel multi-platform profiling methodology. The research initiates by identifying key "Points of Interest" (POIs) from Wikipedia and Wikidata—including politicians and activist organizations—to establish a baseline of influential actors. By mapping these entities to their X (Twitter) profiles, the system captures their follower networks to identify broader ideological communities. Utilizing BERTopic for dynamic topic modeling and longitudinal sentiment analysis, the study deconstructs the central narratives and emotional trajectories of both supporters and opponents. This structured approach moves beyond generic data scraping, offering a refined look at how elite-driven discourse trickles down to public engagement, reflecting the deep societal fractures and persuasive strategies within the Israeli digital sphere.

Keywords: Digital Polarization, Entity-Centric Seeding, Judicial Reform Discourse.

1. Introduction:

In 2023, the Israeli government's proposed changes to the judicial system triggered large-scale demonstrations and an intense public debate in Israel and abroad [1].

The public discourse examined in this study focuses on social media, with an emphasis on X (formerly Twitter) [2]. According to a 2024 Israel Democracy Institute report [3], Facebook was used as a primary or secondary news source by 72% of respondents, Instagram by 46.5%, and X by 12%.

X is particularly useful for computational study because posts by political figures, journalists, organisations, and ordinary users are publicly connected through follower relationships and visible interaction patterns. These posts contain both factual reporting and personal evaluation, making the platform relevant for examining political discourse.

A recent study of toxic behaviour on Twitter/X analysed approximately 1 million tweets from 12,481 panel users together with roughly 6 million tweets from 713,231 accounts they followed [6], illustrating the scale at which behavioural and network-linked discourse can be studied.

The continuous production of posts also creates a methodological challenge: the volume of content grows rapidly, and broad collection strategies can mix politically relevant users with large numbers of incidental or weakly related accounts.

Prior work has linked social-media affordances to information overload [7] and has documented echo chambers and ideologically homogeneous communities across platforms [8]. These findings motivate methods that combine careful population construction with content analysis rather than relying only on raw post volume.

For the judicial-reform debate, this means that identifying who belongs to the relevant discourse population is an important step before interpreting sentiment, emotion, or topics.

Affective polarization is commonly defined in terms of negative evaluations of political out-groups. Druckman, Green, and Iyengar review its relationship with democratic functioning and conclude that evidence for a direct causal link to democratic backsliding in the United States is limited [9].

Separately, Zoizner and Levy report that exposure to toxic behaviour from accounts in users' online networks is associated with adoption of similar toxic behaviour [6].

These findings support studying both the social connections around political actors and the language produced by the resulting user population, while avoiding causal claims that the present observational design cannot establish.

Accordingly, this study builds a computational framework to identify a reform-related discourse population on X and to characterise its posts using supervised classification, sentiment analysis, emotion detection, and topic modelling. The current analysis does not yet separate supporters from opponents, so stance-specific polarization measures remain future work.

2. Related Work:

2.1. Twitter/X as a Platform for Political Discourse

Social media platforms have become central arenas for political communication, enabling rapid spread of information, opinions, and mobilization efforts. Among these platforms, X occupies a distinctive role due to its open structure, real time information flow, and strong emphasis on public interaction through retweets, replies, and hashtags. Although X is not the most widely used platform for news consumption in Israel according to national surveys[3], it remains a highly influential space for political public figures, journalists, activists, and opinion leaders. This asymmetry allows X to function as a key amplifier of political narratives that are often reflected in broader media ecosystems.

Prior research has shown that the interaction mechanisms of social media platforms contribute to information overload and fragmented attention, complicating users' ability to critically assess political information [7]. Moreover, the structural properties of these platforms facilitate the formation of ideologically homogeneous communities, commonly referred to as echo chambers, where exposure to opposing viewpoints is limited[8]. These characteristics make X a particularly valuable platform for studying political discourse dynamics, as it combines high visibility of political content with clear interaction networks that can be systematically analyzed.

2.2. Sentiment Analysis and NLP Methods on Twitter

Natural Language Processing (NLP) techniques have been widely applied to X data to extract sentiment, emotions, and opinions from short, informal texts. Early studies demonstrated the feasibility of using NLP methods to infer user interests and attitudes from tweets, despite challenges related to noise, brevity, and linguistic variation[10]. Subsequent research has applied supervised and unsupervised learning techniques to classify sentiment polarity, detect emotional cues, and analyze public opinion across diverse domains, including public health, consumer behavior, and political communication.[11,12,13-15].

Recent work has increasingly adopted deep learning approaches and distributed word representations, such as Word2Vec and GloVe, to improve sentiment classification performance on X data[16,17]. These models enable richer semantic representations and better generalization across topics. Although many sentiment analysis studies focus on non-political domains, such as finance or consumer services[17,18], the underlying methodologies are directly applicable to political discourse. Collectively, this body of work establishes sentiment analysis as a robust tool for capturing emotional dynamics and evaluative stances within large scale X conversations.

2.3. Topic Modeling and Narrative Patterns

Beyond sentiment polarity, understanding political discourse requires identifying dominant topics and narrative structures that evolve over time. Topic modeling techniques have been widely used to uncover latent themes within large X corpora, enabling researchers to track shifts in public attention and framing across extended periods[12]. Early NLP based approaches demonstrated that user interests and thematic preferences could be inferred from textual patterns, providing a foundation for narrative level analysis[10].

More recent studies have applied topic modeling and thematic extraction to politically charged events, such as public health controversies and mass protests, revealing how competing narratives emerge and interact within online discourse[12,19]. In parallel, research on extremist and radical content has shown that narrative patterns can be critical indicators of ideological positioning and mobilization strategies[20]. These findings highlight the importance of narrative analysis as a complement to sentiment analysis, allowing researchers to move beyond emotional tone and toward a more structural understanding of political discourse online.

2.4. Network Analysis and Online Polarization

Political polarization on social media is deeply intertwined with the structure of online social networks. Network based studies have demonstrated that X users tend to form ideologically aligned clusters, characterized by dense within-group connections and sparse between-group interactions[21]. Such clustering reflects homophily driven dynamics, where users preferentially engage with like minded accounts, reinforcing ideological boundaries[22]. These patterns have been observed across multiple national contexts, including studies of political polarization in India and other democracies[23].

Systematic reviews indicate that echo chambers and ideological segregation are persistent features of online political communication [8]. Affective polarization is also relevant to democratic functioning, although the evidence for a direct causal link to democratic backsliding is not straightforward [9]. Network-linked behavioural research further shows that users can adopt toxic language patterns observed among accounts they follow [6]. Together, these findings motivate analysing social connectivity and textual behaviour jointly, while distinguishing association from causation.

2.5. Gaps in the Literature

Despite extensive research on sentiment analysis, topic modeling, and network based polarization, several gaps remain in the existing literature. First, many studies focus on either textual content or network structure in isolation, limiting their ability to capture the interplay between narrative formation and social connectivity[8,20,22]. Second, while large scale analysis of political polarization on X are well established in the context of the United States and other major democracies[21,23], computational studies addressing the Israeli political context, particularly in relation to the judicial reform, remain relatively scarce.

Existing research on Israel′s judicial reform and constitutional crisis has primarily been conducted from legal, political, and sociological perspectives [1,24,25-27]. This study contributes a computational pipeline that combines entity-centric network expansion, supervised identification of a relevant discourse population, and NLP-based analysis of that population′s posts. Because stance is not yet classified, the current contribution concerns the debate as a whole rather than a direct supporter-versus-opponent comparison.

3. Methodology:

The implemented pipeline consists of four main stages: construction of a verified POI seed set, expansion to a broader candidate population through X follower/following connections, supervised classification with active learning, and text analysis of a sampled high-confidence target population.

All methodological claims in this section refer to artefacts present in the project archive. The final population-selection model operates on the target_population label; the locals_vs_diaspora and person_vs_organisation tasks are evaluated as auxiliary account-characterisation tasks but are not used to filter the final 1,718-account target set.

3.1 POI Identification

Three Hebrew Wikipedia pages related to the reform were used as the initial source: the protest against the judicial reform, the judicial reform in Israel (2023), and support for the judicial reform. Their interlinks produced 825 candidate entities. Manual relevance filtering retained 265 entities: 200 individuals and 65 organisations.

The POI list was then augmented with 147 current and former Knesset members. The combined relevant set therefore contained 412 entities (347 individuals and 65 organisations).

3.2 X Account Mapping and POI Metadata

Relevant entities were manually mapped to X usernames where an active account could be identified. In total, 295 of the 412 relevant entities had an active X account. The archive also contains account-level POI metadata collected from X, including profile text and available follower, following, post-count, and profile fields.

3.3 Candidate Network Expansion

Follower/following collection was available for 264 POI X accounts. The resulting file contains 37,849 connection records across follower, verified-follower, and following relations. Consolidating the connected usernames produced 15,747 unique candidate accounts.

3.4 Candidate Metadata

Candidate metadata include username, display name, biography, location, follower and following counts, verification status, connection location, username-change information, and VPN-related metadata where available. The join-date field is present in the file structure but empty for the candidate pool, and post count is not available as a candidate feature.

3.5 Annotation Scheme and Manual Labeling

Three labels were assigned independently: target_population ({target, non_target, unknown}), locals_vs_diaspora ({local, diaspora, unknown}), and person_vs_organisation ({person, organisation, unknown}). A random seed of 100 candidates was labelled through an interactive HTML tool using profile and recent-content information, with ambiguous cases assigned unknown. Shared decision flowcharts were used to make the labelling procedure consistent.

3.6 Feature Engineering

The classifier pipeline defines seven TF-IDF text representations from biography, username, and full name (each field alone, the three pairwise combinations, and the full concatenation), plus a numeric feature set and a combined TFIDF_all+numeric set, for nine feature sets in total. TF-IDF is fitted directly to the original profile text; profile fields are not translated before classification. Ten numeric/derived features are used: followers, following, follower-to-following ratio, biography length, name length, biography presence, verification, VPN indicator, username-change count, and an indicator for connection via Israel. Identity-only username/full-name feature sets are excluded from deployment, although TFIDF_all and TFIDF_all+numeric still contain those fields; this residual leakage risk is addressed in the limitations.

3.7 Classifier Training and Validation

Six classifiers were trained: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, XGBoost, and AdaBoost. Experiments covered five-fold cross-validation and LOOCV, balanced and unbalanced training, two-class and three-class formulations, and the nine feature sets. Accuracy, precision, recall, F1, and ROC-AUC were logged for each run. Across the three labelled-set sizes, 3,672 experiment rows were recorded.

3.8 Active Learning

The target_population task was designated as the primary task for pool-based uncertainty sampling. After fitting an eligible two-class model, uncertainty was computed as 1 - max(class probability), and the 100 most uncertain unlabelled accounts were exported for manual review. The initial 100-account random seed was followed by two uncertainty-selected batches, producing cumulative labelled sets of 100, 200, and 300 accounts. Selected accounts were labelled for all three target variables.

3.9 Stopping and Confidence Thresholding

The project includes an operational performance summary for the three labelled-set sizes and a separate final confidence-validation stage. The supplied summary reports mean accuracy of 0.657, 0.684, and 0.685 for 100, 200, and 300 labels respectively. Because active-learning batches are deliberately enriched for difficult cases and the archive does

not contain the script that generated this particular filtered summary, the iteration curve is treated as an operational diagnostic rather than a formal convergence estimate. Final target-population membership was instead assessed with a 0.80 confidence threshold and manual validation of a random sample of high-confidence predicted targets.

3.10 Timeline Sampling and Collection

A random sample of 150 accounts was drawn from the high-confidence target-population file for timeline collection. Usable timeline files were obtained for 94 accounts. The scraper configuration capped collection at 300 posts per account; 38 of the 94 collected accounts reached this cap. The timeline files retain tweet identifier, timestamp, username, text, and available interaction counts.

3.11 Translation, Sentiment, and Emotion

Non-empty posts were merged into a 15,858-row corpus. GoogleTranslator (via deep-translator) was used to attempt translation to English, with the original text retained when translation raised an exception. PySentimiento English-language models were then applied for three-way sentiment classification (positive, neutral, negative) and emotion classification (joy, anger, sadness, fear, surprise, disgust, or others).

3.12 Topic Modeling

For BERTopic, URLs, mentions, and retweet markers were removed and posts shorter than five words after cleaning were discarded, leaving 13,720 documents. The model combines sentence embeddings, UMAP, HDBSCAN, and c-TF-IDF. Six configurations were evaluated over minimum topic size {40, 60, 80} and UMAP neighbourhood size {10, 15}, recording topic count and raw outlier share. No topic-coherence score is computed by the archived script. Because none of the tested configurations met its preferred outlier criterion, the script's anti-collapse fallback selected the configuration with the largest number of topics (minimum topic size 40, neighbourhood size 10), after which c-TF-IDF outlier reduction was applied.

3.13 Interpretive Topic Grouping and Scope

BERTopic returned 46 interpreted topics. For report-level exploration, these topic labels were grouped manually into four broad thematic families to compare sentiment and emotion across political and non-political content. The archive does not contain a separate topic-family mapping file, so the family-level analysis should be treated as an interpretive summary rather than an independently reproducible model stage. Named-entity recognition was not implemented, and supporter-versus-opponent stance classification remains outside the current pipeline.

3.14 Implementation Provenance

The archived implementation used for this report contains no Wikidata-enrichment stage and no independent keyword-based X expansion output for the POI set. Those procedures are therefore not claimed as completed components of the reported analysis.

4. Results

The results below report outputs that can be traced to the archived project files. The analysis first describes POI coverage and the candidate population, then evaluates account classifiers and target-population selection, and finally reports sentiment, emotion, and topic results for the collected timeline sample.

Interlink analysis of the three Wikipedia pages yielded 825 candidates, of which 265 were manually retained as relevant (200 individuals and 65 organisations). Adding 147 current and former Knesset members produced 412 relevant entities in total. Of these, 295 had active X accounts: 245 individuals and 50 organisations.

4.1 Entity Identification and X Coverage

Interlink analysis of three Hebrew Wikipedia pages on the reform produced 825 candidate entities, reduced to 265 relevant actors (200 individuals, 65 organisations) after manual filtering; of these, 111 individuals and 50 organisations had active X accounts. Augmenting the set with all 147 current and former Knesset members (134 with active accounts) yielded 412 relevant entities in total, 295 of which maintain an active X presence. Figure 1 summarises this coverage by entity type.

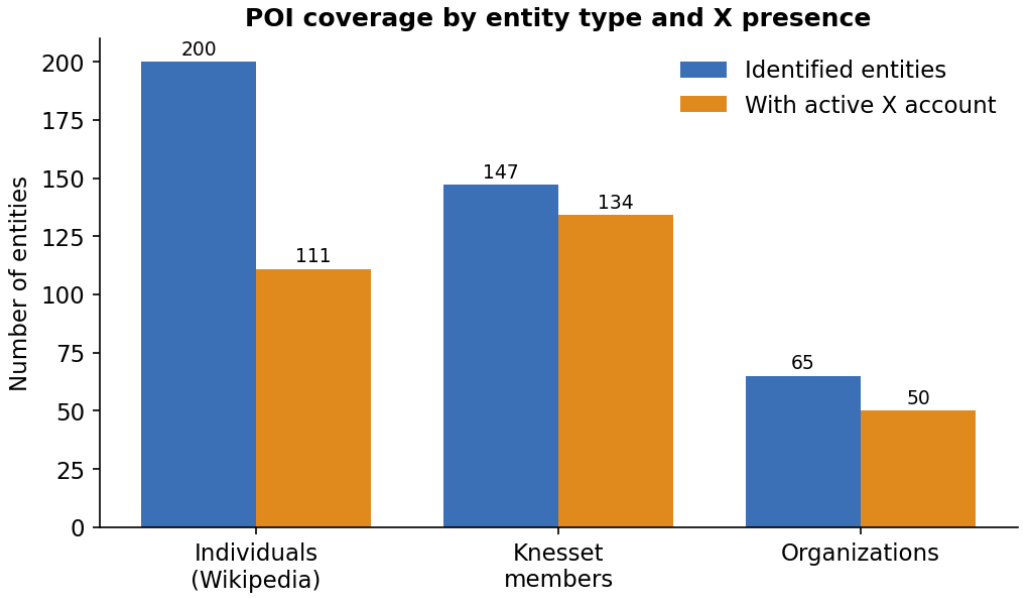


Figure 1. POI coverage by entity type and X presence.

Table 1. Point-of-interest identification and X-account coverage.

Source / type	Entities	With active X account	Coverage
Wikipedia individuals	200	111	55.5%
Wikipedia organisations	65	50	76.9%
Knesset members	147	134	91.2%
Total	412	295	71.6%

4.2 Candidate Population

Follower/following collection from 264 POI accounts produced 37,849 connection records and 15,747 unique candidate usernames. Follower-count metadata are available for 14,662 candidates. The distribution is extremely right-skewed: the median is 292 followers, while the mean is 200,963 because a small number of very high-reach accounts strongly inflate the average. Median following is 548.5, and 4,277 accounts (27.2% of the full candidate file) are marked verified. Figure 2 therefore shows the full available follower distribution on a logarithmic scale.

Table 2. Descriptive statistics of the full candidate pool.

Metric	Value
Connection edges collected	37,849
Unique candidate accounts	15,747
Median followers	292
Mean followers	200,963
Median following	548.5
Verified accounts	4,277 (27.2%)

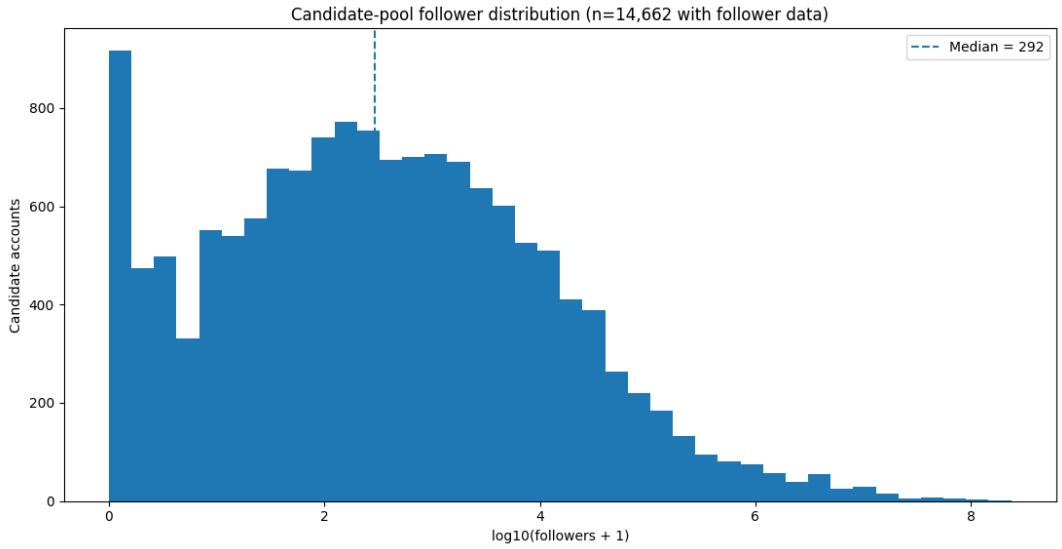


Figure 2. Follower-count distribution of the full candidate pool (log scale; accounts with follower data).

4.3 Manual Annotation and Class Balance

An initial seed of 100 candidates was manually annotated for all three target variables and subsequently expanded over the active-learning iterations. The class distribution of the seed set (Table 3) shows the expected imbalance: most accounts are individuals rather than organisations, and the in-group / out-group split is uneven, motivating the balanced training configurations reported below.

Table 3. Class distribution of the 100-record seed annotation set.

Target variable	Positive	Negative	Unknown
target_population (target / non_target)	34	65	1
locals_vs_diaspora (local / diaspora)	51	39	10
person_vs_organisation (person / org.)	86	10	4

4.4 Classifier Performance

Across the three tasks, two validation strategies, two balancing modes, two class formulations, six classifiers, and nine feature sets, 3,672 experiments were logged. To make the headline comparison reproducible, Table 4 uses iteration 3, the two-class formulation, balanced training, five-fold cross-validation, and excludes the three identity-only feature sets (username only, full name only, and username+full-name). Under that slice, the highest-AUC target_population configuration is Random Forest with TFIDF_all+numeric (accuracy 0.776, F1 0.776, AUC 0.850); locals_vs_diaspora is best with Logistic Regression (0.938, 0.937, 0.972); and person_vs_organisation is best with Random Forest

(0.744, 0.741, 0.888). Because TFIDF_all+numeric still contains username and full name alongside biography text, these scores should be interpreted with a residual identity-leakage caveat. Averaged over the same evaluation slice, Logistic Regression has the highest mean accuracy (0.679), while SVM has the lowest (0.619).

Table 4. Best iteration-3 classifier per task (5-fold CV, balanced, two-class; identity-only feature sets excluded).

Task	Best model	Features	Acc.	F1	AUC
target_population	Random Forest	TF-IDF all + numeric	0.776	0.776	0.850
locals_vs_diaspora	Logistic Reg.	TF-IDF all + numeric	0.938	0.937	0.972
person_vs_organisation	Random Forest	TF-IDF all + numeric	0.744	0.741	0.888

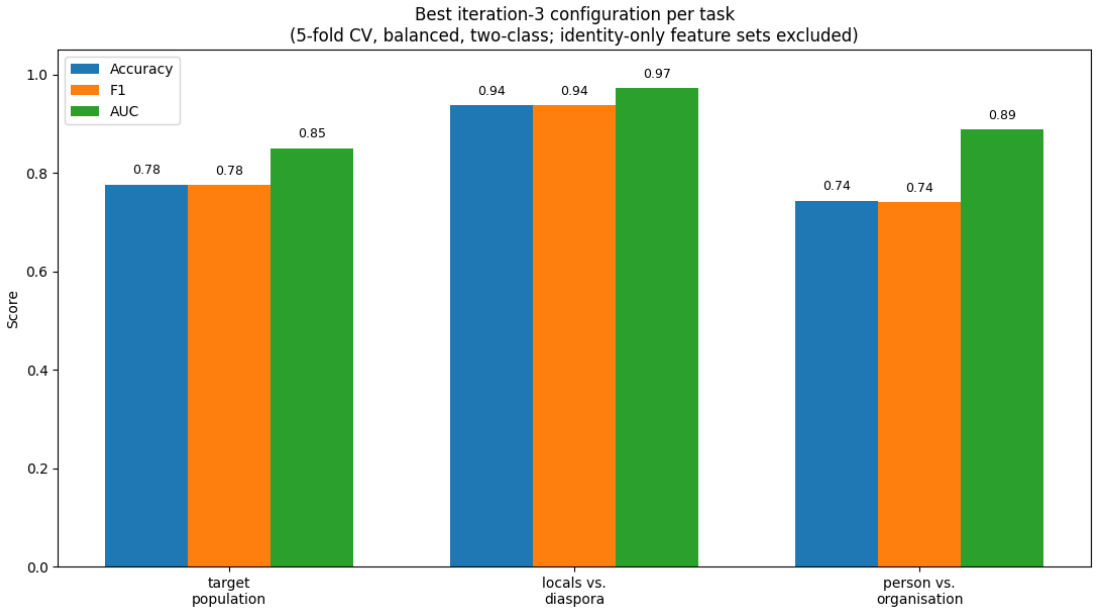


Figure 3. Best-model accuracy, F1 and AUC on the evaluation slice used in Table 4.

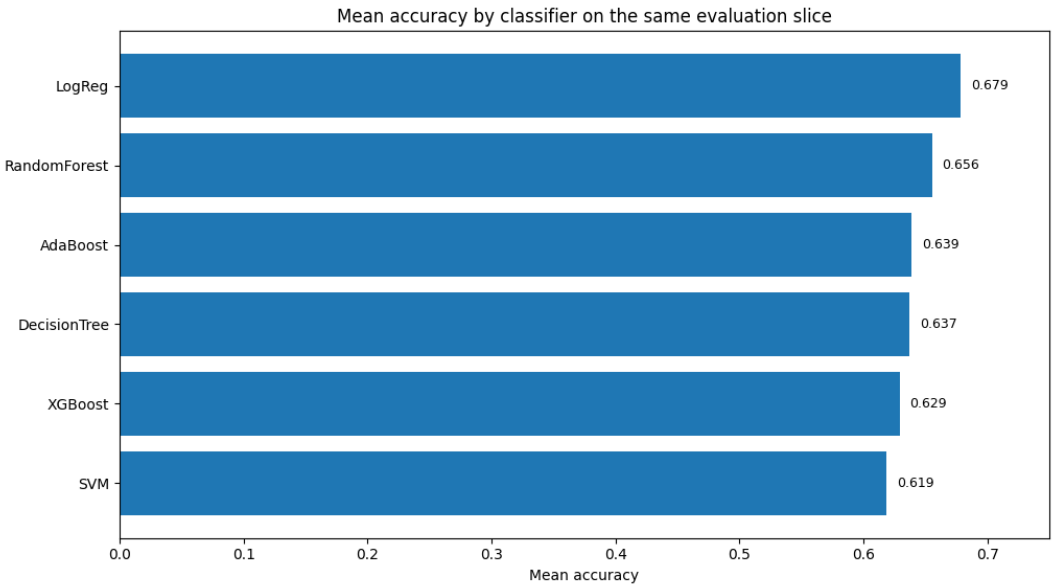


Figure 4. Mean accuracy by classifier on the same evaluation slice as Table 4.

4.5 Active Learning and Stopping Criteria

Three training/evaluation rounds were completed on cumulative labelled sets of 100, 200, and 300 accounts. The move from 100 to 200 and from 200 to 300 used uncertainty-selected additions; the initial 100 accounts were a random seed. The project's performance summary reports mean accuracy of 0.657, 0.684, and 0.685 and mean AUC of 0.642, 0.657, and 0.651 (Table 5, Figure 5). The final accuracy change from 200 to 300 labels is only +0.001. However, because uncertainty sampling changes the difficulty and composition of each labelled set, and because the archive does not preserve the code that generated this filtered summary, the curve is used here as an operational stopping diagnostic rather than evidence of statistical convergence. Annotation was stopped after 300 labels and the final threshold was evaluated separately by manual validation.

Table 5. Project performance summary by labelled-set size (operational diagnostic).

Iteration	Labelled examples	Mean accuracy	Mean AUC
1	100	0.657	0.642
2	200	0.684	0.657
3	300	0.685	0.651

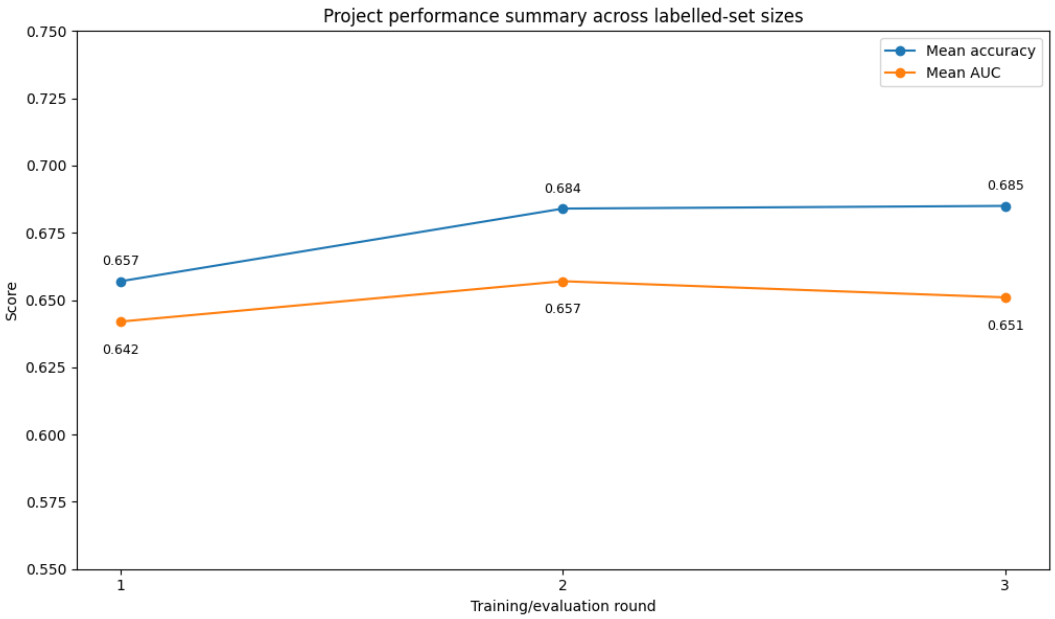


Figure 5. Project performance summary across labelled-set sizes; this is not treated as a formal convergence test.

4.6 Confidence Thresholding and Population Validation

The final target-population scoring file contains 15,447 previously unlabelled users. Median model confidence is approximately 0.77. At a confidence threshold of 0.80, 6,420 users of either predicted class (41.6%) meet the confidence requirement; among them, 1,718 are predicted as target and form the final target-population file. For validation, 40 accounts were sampled randomly from these high-confidence predicted targets and manually checked; 37 were judged to belong to the target population, giving a 92.5% hit rate. This directly validates precision within the sampled high-confidence target set, not recall or representativeness of the wider candidate population.

Table 6. Number of scored users meeting alternative confidence thresholds.

Confidence threshold	All users meeting threshold	Share of scored users
≥ 0.70	10,593	68.6%
≥ 0.80 (selected)	6,420	41.6%
≥ 0.90	2,180	14.1%

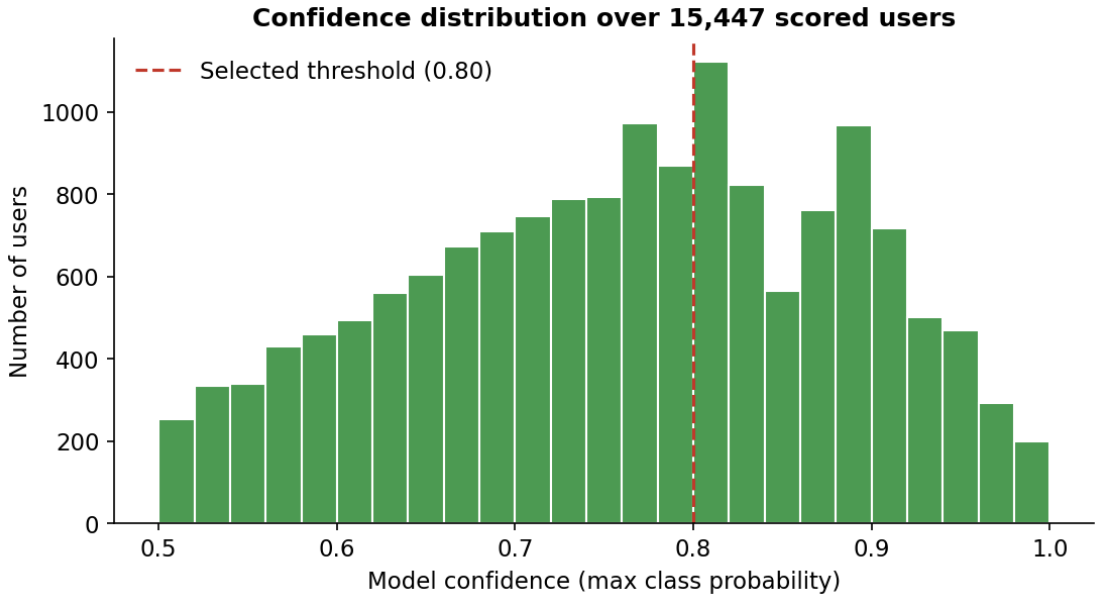


Figure 6. Confidence distribution of the final model over all scored users.

4.7 Target-Population Discourse Corpus

A random sample of 150 high-confidence predicted target accounts was selected for timeline collection, and usable timeline files were obtained for 94 accounts. These files contain 16,338 collected posts, of which 15,858 have non-empty text. The observed period spans 1 January through 6 October 2023 in the stored UTC timestamps. Collection is right-truncated for highly active users: 38 of the 94 accounts (40.4%) reached the 300-post cap. Figures 7 and 8 therefore describe the collected sample rather than population-level posting activity.

Table 7. Summary of the collected discourse corpus.

Metric	Value
Accounts with collected timelines	94
Total tweets collected	16,338
Tweets with non-empty text	15,858
Date range	2023-01-01 to 2023-10-06 (UTC)

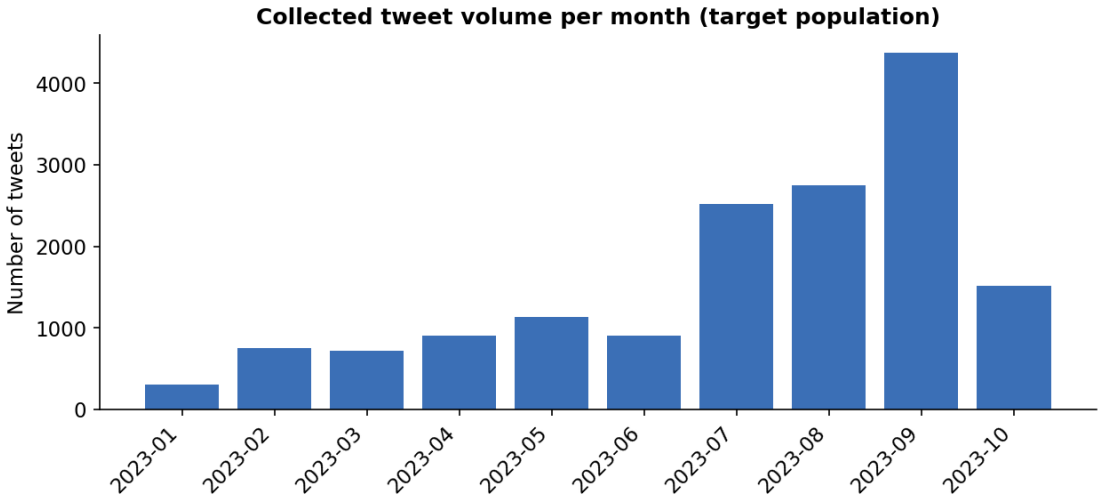


Figure 7. Monthly collected post volume in the 94-account timeline sample.

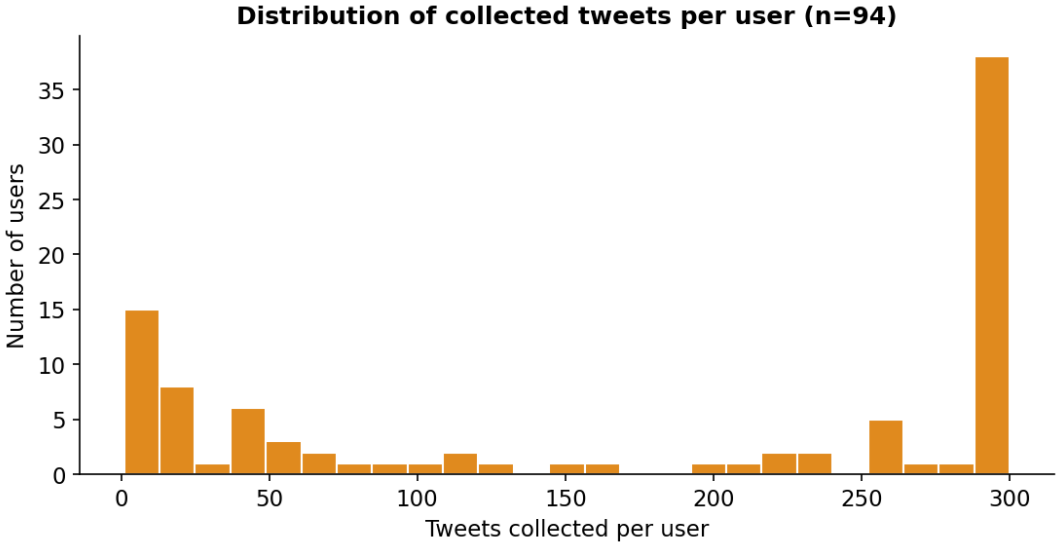


Figure 8. Distribution of collected posts per user in the 94-account sample.

4.8 Sentiment of the Discourse

The translation pipeline attempted to translate all 15,858 non-empty posts to English and retained the original text when translation failed. PySentimiento then assigned each row to positive, neutral, or negative sentiment. The resulting distribution is 37.7% negative, 44.2% neutral, and 18.0% positive, a negative-to-positive ratio of approximately 2.1:1 (Table 8, Figure 9). Across months, the positive share remains around one fifth or lower, rather than exceeding it substantially.

Table 8. Sentiment distribution over the target-population corpus (PySentimiento, n = 15,858).

Sentiment	Tweets	Share
Neutral	7,010	44.2%
Negative	5,986	37.7%
Positive	2,862	18.0%
Total	15,858	100.0%

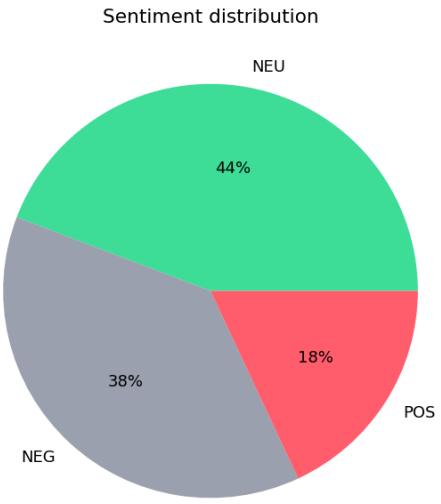


Figure 9. Sentiment distribution of the target-population corpus.

The monthly breakdown (Figure 10) shows 45.0% negative sentiment in February 2023 and 41.9% in July. During April to June, the negative share is approximately 33-37% and the positive share is around 20%. These changes coincide with different phases of the reform debate, but the timeline sample is incomplete and capped, so the temporal pattern should be interpreted descriptively rather than as evidence that political events caused the observed changes.

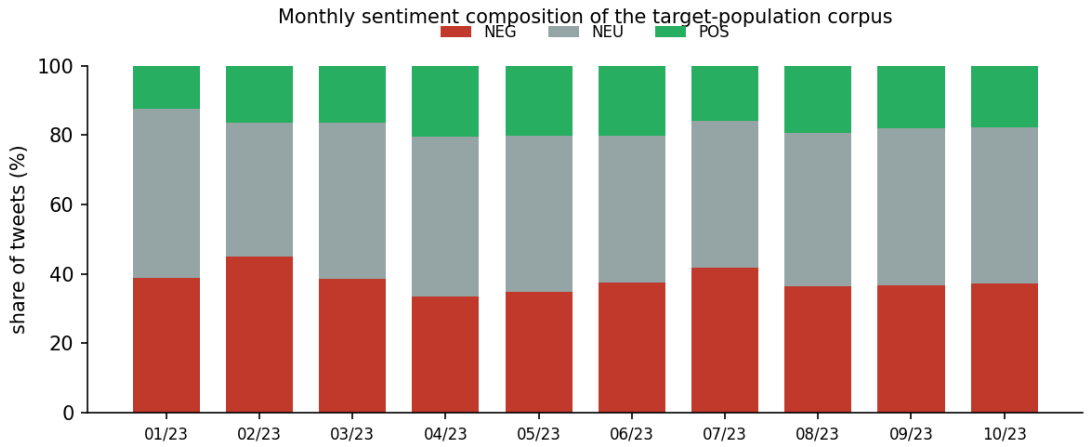


Figure 10. Monthly sentiment composition of the corpus, January to October 2023.

4.9 Emotion Profile

The PySentimiento emotion analyser assigns one of six specific emotions or the label 'others' when no listed emotion is selected. In the corpus, 61.3% of posts are labelled 'others'. Among specific labels, disgust is most frequent at 20.1%, followed by joy (8.1%), anger (4.5%), sadness (2.8%), fear (2.2%), and surprise (1.0%) (Table 9, Figure 11).

Table 9. Emotion distribution over the target-population corpus (PySentimiento, n = 15,858).

Emotion	Tweets	Share
Others (no dominant emotion)	9,724	61.3%
Disgust	3,189	20.1%
Joy	1,286	8.1%
Anger	715	4.5%
Sadness	439	2.8%
Fear	344	2.2%
Surprise	161	1.0%

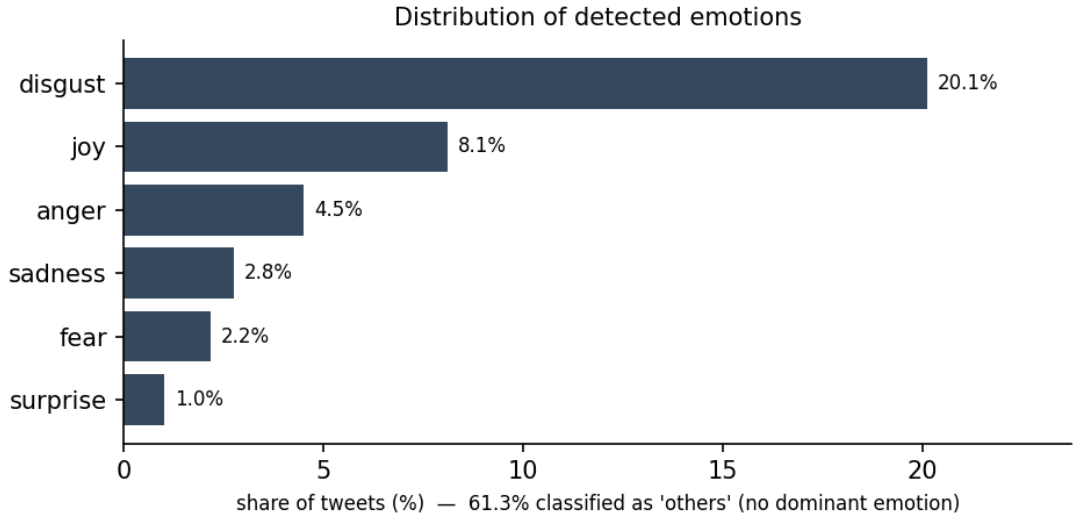


Figure 11. Distribution of detected emotions (the 'others' class is omitted).

Disgust is therefore the dominant specific model label in this translated corpus, substantially exceeding anger and fear. This result should not be equated directly with psychological contempt, moral rejection, or hostility toward a particular political camp: the classifier predicts an emotion category from text but does not identify the target or cause of that emotion. Longitudinally (Figure 12), the disgust share is 23.8% in January, 16.1% in April, 17.3% in May, 22.4% in July, and 23.3% in the final partial month, while anger reaches 11.9% in February before returning to lower levels. These are descriptive model outputs rather than causal behavioural findings.

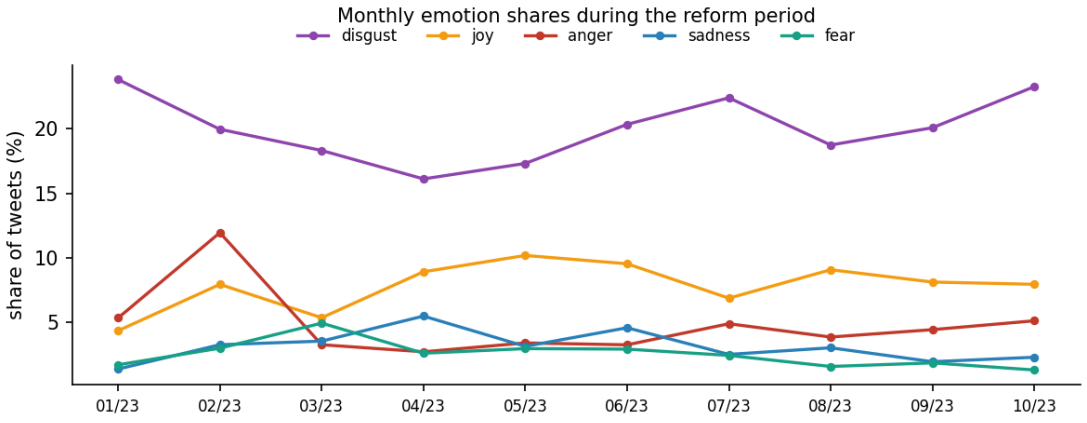


Figure 12. Monthly emotion shares during the reform period.

4.10 Topic Structure of the Discourse

Topics were extracted with BERTopic after removing URLs, mentions, and retweet markers and discarding cleaned posts shorter than five words, leaving 13,720 documents. English stop words were removed at vectorisation. The archived grid-search script records

only the number of recovered topics and raw outlier share; it does not compute topic coherence.

Six configurations were compared over minimum topic size {40, 60, 80} and UMAP neighbourhood size {10, 15}. None met the script's preferred raw-outlier threshold of 40%, so its anti-collapse fallback selected the configuration with the most topics: minimum topic size 40 and neighbourhood size 10, which produced 46 topics with 54.8% raw outliers. BERTopic's c-TF-IDF outlier-reduction step then reassigned most outliers, leaving 31 documents (0.2% of the 13,720 modelled documents) with topic -1.

Table 10. BERTopic grid search over minimum topic size and neighbourhood size.

Min. topic size	Neighbours	Topics recovered	Outlier share
40	10	46 (selected)	54.8%
40	15	31	52.4%
60	10	25	60.1%
60	15	23	60.5%
80	10	18	61.6%
80	15	17	56.0%

The 46 recovered topics divide into a political core and a substantial everyday remainder. The largest single cluster is a greetings-and-thanks topic (1,066 documents), followed by family/personal life (923) and a broad 'State of Israel' topic (903). Explicitly political clusters appear immediately below these: the Prime Minister and the criminal proceedings against him (564), Netanyahu as a named subject (430), the High Court and its judges (300), democracy versus dictatorship (274), protest and its leaders (252), the reasonableness standard and Basic Laws (187), and the Kaplan Street demonstrations (124). A parallel security-and-conflict family covers the IDF and its commanders (468), Gaza (389), Lebanon and Hezbollah (147) and casualty reporting (190). Table 11 lists the twelve largest topics and Figure 13 the fifteen largest.

Table 11. The twelve largest BERTopic topics.

ID	Interpreted label	Leading terms	Tweets
T0	Greetings and thanks	thank, happy, year, dear	1,066
T2	Family and personal life	mother, like, really, woman	923

ID	Interpreted label	Leading terms	Tweets
T1	The State of Israel	israel, state, israeli	903
T4	Good-morning routine	morning, good morning, come	603
T6	PM and criminal proceedings	minister, prime minister, criminal	564
T5	Money, taxes, cost of living	nis, money, pay, tax	501
T8	IDF and commanders	idf, army, general, commander	468
T7	Food and daily life	eat, coffee, just, like	463
T10	Obituaries and remembrance	memory, blessed, late	441
T3	Netanyahu	netanyahu, prime minister	430
T9	Gaza	gaza, idf, strip	389
T11	Police and arrests	police, arrested, suspicion, negev	383

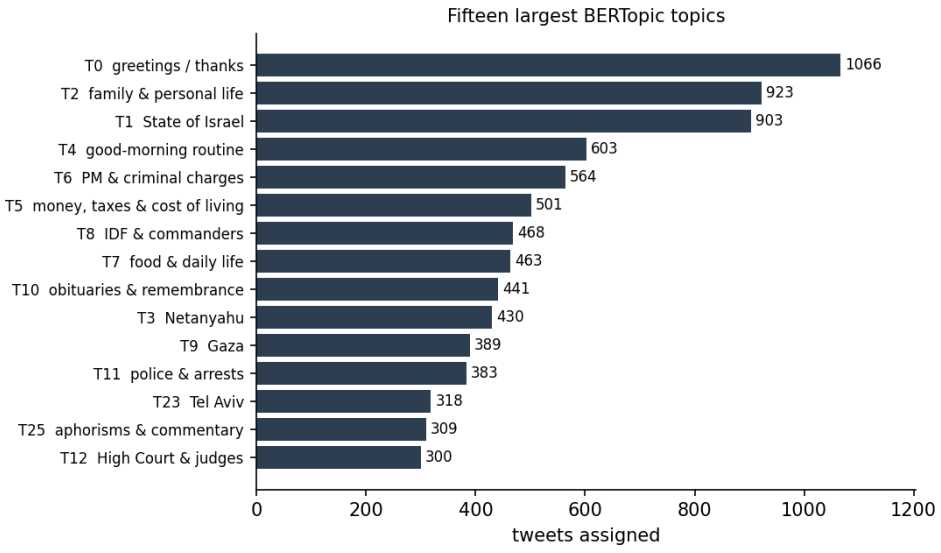


Figure 13. The fifteen largest BERTopic topics by number of assigned tweets.

4.11 Topic-Conditional Sentiment and Emotion

For an exploratory report-level comparison, the 46 interpreted topic labels were manually grouped into four thematic families. After excluding the 31 documents that remained unassigned, 13,689 posts have one of the 46 topic assignments. The reported family

grouping places 32.9% of these posts in reform/politics/protest topics and 48.3% in everyday/non-political topics. Within the reported grouping, negative sentiment is 51.5% in the political family versus 34.6% in the everyday family, and the disgust label is 32.4% versus 16.5%. Because the archive does not contain the topic-to-family mapping as a separate data artefact, these family-level comparisons should be treated as interpretive summaries rather than a fully reproducible model output.

Table 12. Sentiment and emotion by manually interpreted thematic group (46 topics; 13,689 assigned posts).

Thematic group	Topics	Share of corpus	Negative	Positive	Disgust
Reform, politics and protest	17	32.9%	51.5%	10.3%	32.4%
Security and conflict	6	11.2%	43.9%	8.0%	17.8%
Religion and societal cleavages	6	7.6%	37.9%	16.1%	25.1%
Everyday and non-political	17	48.3%	34.6%	23.7%	16.5%

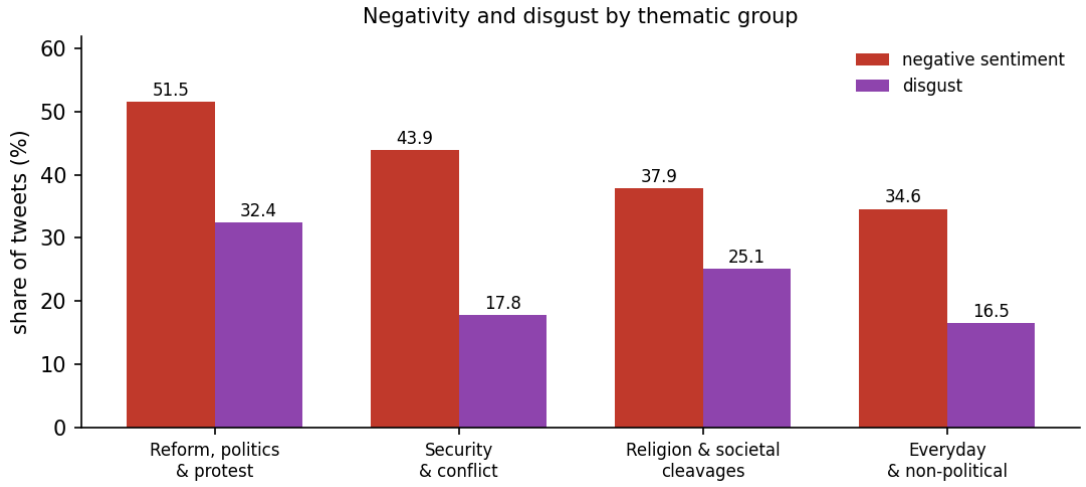


Figure 14. Negative-sentiment and disgust shares by thematic group.

At the individual-topic level, the archived topic assignments support the contrasts shown in Table 13: casualty reporting is 67.9% negative; the Prime Minister/criminal-proceedings topic is 63.7% negative and 48.9% disgust; police/arrests is 62.4% negative; accusations of lying is 61.6% negative and 44.1% disgust; and democracy-versus-dictatorship is 59.1% negative. At the opposite end, songs/music, good-morning routine, and Shabbat greetings are 15.0%, 16.7%, and 17.8% negative respectively. These aggregate differences are consistent with an association between topic and tone, but they do not by themselves

establish a within-user causal effect or rule out differences in which users contribute to each topic.

Table 13. The five most negative and three least negative individual topics.

ID	Interpreted label	Tweets	Negative	Disgust
T34	Casualties and violent incidents	190	67.9%	26.3%
T6	PM and criminal proceedings	564	63.7%	48.9%
T11	Police and arrests	383	62.4%	19.6%
T42	Accusations of lying	177	61.6%	44.1%
T29	Democracy vs. dictatorship	274	59.1%	27.0%
T22	Shabbat greetings	169	17.8%	9.5%
T4	Good-morning routine	603	16.7%	3.8%
T20	Songs and music	227	15.0%	5.7%

5. Discussion

Taken together, the results support four descriptive findings and methodological observations about the 2023 judicial-reform discourse sample.

5.1 Entity-Centric Seeding Produces a High-Precision Target Set

The pipeline starts from 412 manually verified relevant entities, 295 of which had active X accounts, and expands through connection data collected for 264 of those accounts. This produces a broad and noisy pool of 15,747 candidate usernames. A classifier and 0.80 confidence threshold reduce that pool to 1,718 predicted target accounts, and a random manual check of 40 high-confidence predicted targets yields 92.5% precision in that sample. This supports the usefulness of entity-centric seeding for finding relevant accounts, but it does not establish recall or representativeness: relevant users disconnected from the chosen POIs can be missed, and incidental followers can enter the candidate pool.

5.2 Negativity Is Associated with Political Topics

The corpus-level sentiment distribution is 37.7% negative and 18.0% positive. In the report's manual thematic grouping, reform/politics/protest topics are 51.5% negative, while everyday/non-political topics are 34.6% negative; several explicitly political individual topics exceed 59% negative, whereas routine greetings and music are below 18%. This pattern is consistent with topic-conditioned differences in tone. It should not be described

as proof that political subject matter 'switches on' hostility within the same individuals, because no paired within-user or mixed-effects statistical model was fitted and topic participation varies by user.

5.3 Disgust Is the Dominant Specific Emotion Label

The emotion model assigns disgust to 20.1% of posts, compared with 4.5% anger and 2.2% fear. The result is notable as a model-based description of the language in the translated corpus, but 'disgust' should not be converted automatically into claims about contempt, delegitimisation, or the identity of an emotional target. The broader literature also warrants caution: Druckman, Green, and Iyengar find limited evidence for a direct causal link between affective polarization and democratic backsliding in the United States [9]. The present observational analysis therefore treats emotion labels as descriptive features rather than evidence of democratic effects.

5.4 The Sample Contains Both Political and Everyday Speech

The report-level thematic grouping assigns almost half of the topic-modelled posts to everyday and non-political themes. This shows that the sampled accounts are not uniformly political in what they post and that full timeline collection can provide a non-political comparison set. Because the thematic families are manually interpreted and the sample contains only 94 collected accounts, this finding should be understood as a property of the present corpus rather than as a general estimate of how all reform-discourse users divide their attention.

6. Conclusions

This project builds an end-to-end computational pipeline for identifying a population connected to the 2023 Israeli judicial-reform debate on X and characterising a sample of its posts. The implemented workflow covers POI identification, follower/following expansion, manual labelling, supervised classification with active learning, confidence-based population selection, timeline collection, translation, sentiment and emotion analysis, and BERTopic topic modelling.

The verified outputs are substantial: 412 relevant POIs, 295 active X accounts, connection data from 264 POI accounts, 37,849 connection records, and 15,747 unique candidate users. The experiment log contains 3,672 classifier runs. On the reproducible iteration-3, balanced, two-class, five-fold-CV slice used in Table 4, the highest AUCs are 0.850 for target_population, 0.972 for locals_vs_diaspora, and 0.888 for person_vs_organisation. Two uncertainty-sampling acquisitions expanded the initial random 100-account seed to 300 cumulative labels. Applying a 0.80 confidence threshold to the final target-population scores yielded 1,718 predicted target accounts, with 37 of 40 randomly checked high-confidence predictions validated as target (92.5%).

Timeline collection yielded 16,338 posts from 94 sampled accounts, including 15,858 non-empty texts. PySentimiento labels 37.7% of these posts negative, 44.2% neutral, and 18.0% positive, and disgust is the most frequent specific emotion label at 20.1%. BERTopic produces 46 topics from 13,720 preprocessed documents, with 31 remaining unassigned

after outlier reduction. The report's exploratory thematic grouping indicates more negative and disgust-labelled content in political topics than in everyday topics, but this association is not treated as causal or as a direct measure of polarization between opposing camps.

The main methodological contribution is the entity-centric population-construction pipeline: rather than beginning with a hashtag-only post sample, it builds a candidate user pool around verified actors and then validates target membership with supervised learning and manual checking. The current report characterises the discourse population as a whole. A true supporters-versus-opponents analysis requires an additional stance-classification or community-detection stage and remains future work.

7. Limitations and Future Work

7.1 Limitations

Stance is not separated. The target_population classifier identifies whether an account belongs to the reform-related discourse population, not whether it supports or opposes the reform. The current sentiment, emotion, and topic results therefore characterise the collected debate sample as a whole and cannot be attributed to either political camp.

The text corpus is a partial, capped sample. Although 150 high-confidence target accounts were randomly selected for scraping, usable timelines were obtained for 94. The scraper retained at most 300 posts per account, and 38 of the 94 collected accounts (40.4%) reached that cap. The resulting corpus is not a probability sample of all 1,718 predicted target accounts, and monthly volume should not be interpreted as a population-level activity estimate.

Text analysis depends on machine translation. The pipeline attempts English translation and falls back to the original text on translation errors. Translation can lose sarcasm, Hebrew political slang, legal and religious terminology, and other pragmatic cues that are important for sentiment and emotion classification. Consequently, neutral/'others' rates and specific emotion labels may partly reflect translation and model limitations.

Classifier and annotation limitations remain. The candidate join-date field is empty and post count is unavailable, so account age and lifetime activity are not used. Identity-only username/full-name feature sets are excluded from deployment, but the best combined TFIDF_all+numeric configurations still contain username and full name, leaving a residual identity-leakage risk. The final 300-label set contains only 39 organisation examples in the two-class person-versus-organisation task. Two authors used shared flowcharts for annotation, but no inter-annotator agreement statistic was computed. Named-entity recognition was not implemented.

Topic-family and temporal limitations also apply. The four thematic families are a manual interpretive grouping and the archive does not contain a separate topic-to-family mapping file, limiting exact reproducibility of those aggregate comparisons. The corpus ends at the beginning of 7 October 2023 local time (6 October in the stored UTC timestamps), and the results should not be extrapolated to the substantially different discourse environment after the outbreak of war.

7.2 Future Work

The most important extension is explicit stance classification. A new labelled seed distinguishing supporters and opponents could be expanded with active learning and compared with unsupervised community structure in the existing follower graph. This would enable genuine between-camp measures such as differential topic emphasis, asymmetric sentiment, and cross-camp connectivity. The classifier analysis should also be rerun with username and full-name information removed entirely to quantify how much performance remains when identity leakage is eliminated.

Additional work should analyse Hebrew natively with models such as HeBERT or AlephBERT, implement named-entity recognition, persist a reproducible topic-to-family mapping, expand timeline collection beyond the present 94 accounts and 300-post cap, and fit user-level statistical models (for example, mixed-effects models) to test whether political topics are associated with more negative or disgust-labelled language within the same users rather than only at aggregate topic level.

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